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TOP 111 PHYSICS QUESTIONS

OF

JEE MAIN 2026 JAN ATTEMPT

MOHIT GOENKA

Founder of Eduniti | Alumnus IIT Kharagpur

INTRODUCTION

In **JEE Main 2026 Jan attempt** a total of **250 questions** were asked in physics.

This file contains **top 111 questions** out of those 250 questions selected by Mohit sir. Along with answer key you also get its detailed video solution.

Aspirants are advised to print it and further solve it in **time bound manner**. A maximum of **4 hrs is recommended** for solving these 111 questions

VIDEO SOLUTION TOP 111Q OF JAN 2026

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MOHIT GOENKA
Founder of Eduniti | Alumnus IIT KGP

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U&D, ERRORS & VECTORS

1. Keeping the significant figures in view, the sum of the physical quantities 52.01m, 153.2 m and 0.123 m is :

(1) 205 m (2) 205.333 m

(3) 205.33 m (4) 205.3 m

2. If $\in$, E and t represent the free space permittivity, electric field and time respectively, then the unit of $\frac{\in E}{t}$ will be :

(1) Am (2) Am^2

(3) A/m^2 (4) A/m

3. Four persons measure the length of a rod as 20.00 cm, 19.75 cm, 17.01 cm and 18.25 cm. The relative error in the measurement of average length of the rod is :

(1) 0.24 (2) 0.18

(3) 0.06 (4) 0.08

4. In a screw gauge, the zero of the circular scale lies 3 divisions above the horizontal pitch line when their metallic studs are brought in contact. Using this instrument thickness of a sheet is measured. If pitch scale reading is 1 mm and the circular scale reading is 51 then the correct thickness of the sheet is _____ mm.

[Assume least count is 0.01 mm]

(1) 1.50 (2) 1.48

(3) 1.54 (4) 1.51

5. A ball of radius r and density ρ dropped through a viscous liquid of density σ and viscosity η attains its terminal velocity at time t, given by $t = A\,\rho^a\, r^b\, \eta^c\, \sigma^d$, where A is a constant and a, b c and d are integers. The value of $\frac{b+c}{a+d}$ is _____.

CLICK FOR SOLUTION

6. When both jaws of vernier callipers touch each other, zero mark of the vernier scale is right to zero mark of main scale, 4th mark on vernier scale coincides with certain mark on the main scale. While measuring the length of a cylinder, observer observes 15 divisions on main scale and 5th division of vernier scale coincides with a main scale division. Measured length of cylinder is _____ mm. (Least count of Vernier calliper = 0.1 mm)

(1) 15.4 (2) 15.1

(3) 15.5 (4) 15.9

7. The time period of a simple harmonic oscillator is $T = 2\pi\sqrt{\frac{k}{m}}$. The measured value of mass (m) of the object is 10 g with an accuracy of 10 mg, and time for 50 oscillations of the spring is found to be 60 s using a watch of 2 s resolution. Percentage error in determination of spring constant(k) is _____%.

(1) 3.43 (2) 3.35 (3) 7.60 (4) 6.76

KINEMATICS

8. A river of width 200 m is flowing from west to east with a speed of 18 km/h. A boat, moving with speed of 36 km/h in still water, is made to travel one-round trip (bank to bank of the river). Minimum time taken by the boat for this journey and also the displacement along the river bank are _____ and _____ respectively.

(1) 20 s and 100 m

(2) 40 s and 0 m

(3) 40 s and 200 m

(4) 40 s and 100 m

9. A paratrooper jumps from an aeroplane and opens a parachute after 2 s of free fall and starts deaccelerating with 3 m/s^2. At 10 m height from ground, while descending with the help of parachute, the speed of paratrooper is 5 m/s. The initial height of the aeroplane is _____m.
($g = 10\ m/s^2$)

(1) 62.5 (2) 92.5

(3) 20 (4) 82.5

10. A boy thrown a ball into air at 45° from the horizontal to land it on a roof of a building of height H. If the ball attains maximum height in 2 s and lands on the building in 3 s after launch, then value of H is _____ m.
($g = 10\ m/s^2$)
(1) 20 (2) 10 (3) 25 (4) 15

11. Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begins to fall. The height at which the fourth drop will be from ground, at the instant when the first drop strikes the ground is _____ m.
($g = 10\ m/s^2$)
(1) 2.5 (2) 4.0
(3) 4.2 (4) 3.8

12. A particle starts moving from time t = 0 and its coordinate is given as $x(t) = 4t^3 - 3t$.

A. The particle returns to its original position (origin) 0.866 units later

B. The particle is 1 unit away from origin at its turning point.

C. Acceleration of the particle is non-negative.

D. The particle is 0.5 units away from origin at its turning point.

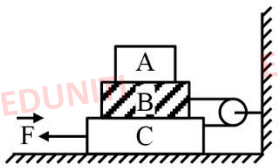
E. Particle never turns back as acceleration is non-negative.

Choose the **correct** answer from the options given below :
(1) A,C,D only (2) A,B,C only
(3) C,E only (4) A,C only

CLICK FOR SOLUTION

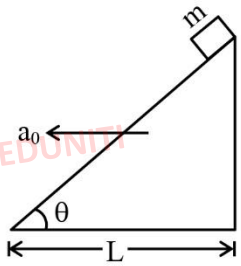
LAWS OF MOTION

13. In the given figure the blocks A, B and C weigh 4 kg, 6 kg and 8 kg respectively. The co-efficient of sliding friction between any two surfaces is 0.5. The force $\vec{F}$ required to slide the block C with constant speed is _____ N. (Used $g = 10\ m/s^2$)



14. A flexible chain of mass *m* hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30°. Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is _____
(1) $\frac{\sqrt{3}}{2}mg$ (2) $\frac{1}{2}mg$ (3) mg (4) $\sqrt{3}mg$

15. A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0. The angle between the inclined plane and ground is θ and its base length is L. Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is _____.



(1) $\sqrt{\frac{2L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

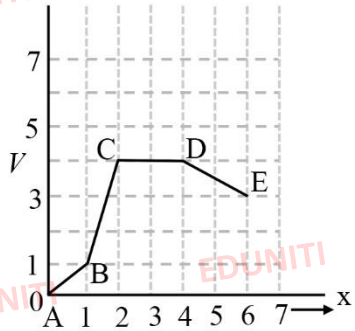
(2) $\sqrt{\frac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

(3) $\sqrt{\frac{4L}{g\cos^2\theta - a_0\sin\theta\cos\theta}}$

(4) $\sqrt{\frac{2L}{g\sin\theta - a_0\cos\theta}}$

WEP & CIRCULAR MOTION

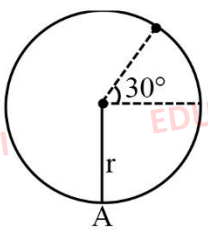
16. Potential energy (V) versus distance (x) is given by the graph, Rank various regions as per the magnitudes of the force (F) acting on a particle from high to low.



(1) $F_{BC} > F_{CD} > F_{DE} > F_{AB}$
(2) $F_{CD} > F_{AB} > F_{BC} > F_{DE}$
(3) $F_{CD} > F_{DE} > F_{AB} > F_{BC}$
(4) $F_{BC} > F_{AB} > F_{DE} > F_{CD}$

17. In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° shown in figure, the velocity at the bottom point (A) of the circular path is

(g = gravitational acceleration)

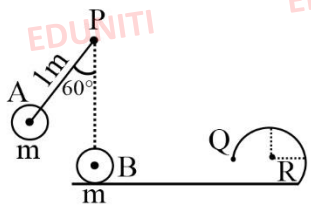


(1) $\sqrt{5gr}$ (2) $\sqrt{\frac{7}{2}gr}$

(3) $\sqrt{4gr}$ (4) $\sqrt{\frac{5}{2}gr}$

COM & COLLISION

18. A small bob A of mass m is attached to a massless rigid rod of length 1m pivoted at point P and kept at an angle of 60° with vertical as shown in figure. At distance of 1m below point P, an identical bob B is kept at rest on a smooth horizontal surface that extends to a circular track of radius R as shown in figure. If bob B just manages to complete the circular path of radius R upto a point Q after being hit elastically by bob A, then radius R is _____m.



(1) $\frac{3}{5}$ (2) $\frac{1}{5}$

(3) $\frac{2+\sqrt{3}}{5}$ (4) $\frac{2-\sqrt{3}}{5}$

19. A body of mass 14 kg initially at rest explodes and breaks into three fragments of masses in the ratio 2 : 2 : 3. The two pieces of equal masses fly off perpendicular to each other with a speed of 18 m/s each. The velocity of the heavier fragment is _____m/s.

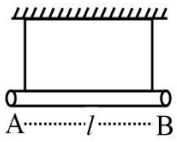
(1) $10\sqrt{2}$ (3) 12

(2) $12\sqrt{2}$ (4) $24\sqrt{2}$

CLICK FOR SOLUTION

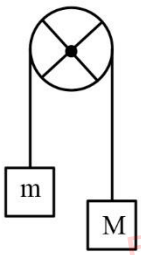
ROTATION

20. A uniform rod of mass m and length l suspended by means of two identical inextensible light strings as shown in figure. Tension in one string immediately after the other string is cut, is _____. (g acceleration due to gravity)



(1) mg/2 (2) mg/4
(3) mg/3 (4) mg

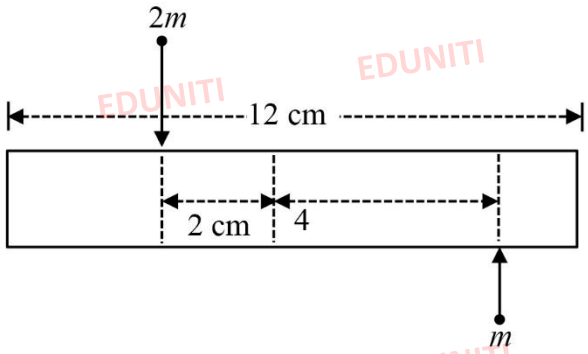
21. The pulley shown in figure is made using a thin rim and two rods of length equal to diameter of the rim. The rim and each rod have a mass of M. Two blocks of mass of M and m are attached to two ends of a light string passing over the pulley, which is hinged to rotate freely in vertical plane about its centre. The magnitudes of the acceleration experienced by the blocks is____ (assume no slipping of string on pulley.)



(1) $\frac{(M-m)g}{\left[\left(\frac{13}{6}\right)M+m\right]}$ (3) $\frac{(M-m)g}{\left[\left(\frac{8}{3}\right)M+m\right]}$

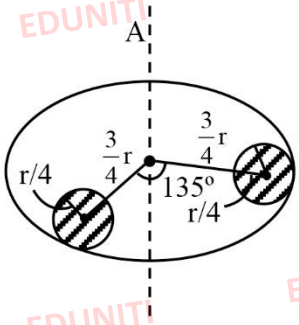
(2) $\frac{(M-m)g}{M+m}$ (4) $\frac{(M-m)g}{2M+m}$

22. A uniform bar of length 12 cm and mass 20 m lies on a smooth horizontal table. Two point masses m and 2 m are moving in opposite directions with same speed of v and in the same plane as the bar, as shown in figure. These masses strike the bar simultaneously and get stuck to it. After collision the entire system is rotating with angular frequency ω. The ratio of v and ω is :



(1) 33 (2) $2\sqrt{88}$
(3) 66 (4) 32

23. Suppose there is a uniform circular disc of mass M kg and radius r m shown in figure. The shaded regions are cut out from the disc. The moment of inertia of the remainder about the axis A of the disc is given by $\frac{x}{256}$ Mr^2. The value of x is _____.

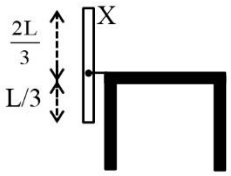


24. Two masses 400 g and 350 g are suspended from the ends of a light string passing over a heavy pulley of radius 2 cm. When released from rest the heavier mass is observed to fall 81 cm in 9 s. The rotational inertia of the pulley is _____ $kg.m^2$. ($g = 9.8\ m/s^2$)

(1) 9.5×10^{-3} (2) 4.75×10^{-3}
(3) 1.86×10^{-2} (4) 8.3×10^{-3}

CLICK FOR SOLUTION

25. A thin uniform rod (X) of mass M and length L is pivoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure. The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is_____. (g = gravitational acceleration)

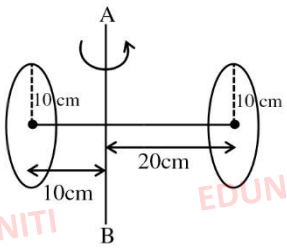


(1) $\sqrt{\frac{3}{2}\frac{g}{L}}$ (2) $\frac{3}{\sqrt{2}}\sqrt{\frac{g}{L}}$

(3) $\frac{1}{\sqrt{2}}\sqrt{\frac{g}{L}}$ (4) $\sqrt{\frac{3g}{L}}$

26. Two circular discs of radius each 10 cm are joined at their centres by a rod of length 30 cm and mass 600 gm as shown in figure.

If the mass of each disc is 600 gm and applied torque between two discs is 43×10^5 dyne cm, the angular acceleration of the discs about the given axis *AB* is _____ rad/s^2.



(1) 22 (2) 11

(3) 100 (4) 27

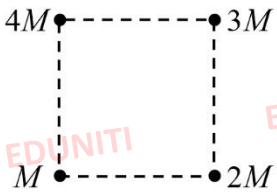
GRAVITATION

27. Initially a satellite of 100 kg is in a circular orbit of radius $1.5R_E$. This satellite can be moved to a circular orbit of radius $3R_E$ by supplying $\alpha \times 10^6$ J of energy. The value of α is _____.
(Take Radius of Earth $R_E = 6 \times 10^6$ m and g = 10 m/s^2)

(1) 150 (2) 500

(3) 100 (4) 1000

28. Net gravitational force at the centre of a square is found to be F_1 when four particles having mass *M, 2M, 3M* and *4M* are placed at the four corners of the square as shown in figure and it is F_2 when the positions of *3M* and *4M* are interchanged. The ratio $\frac{F_1}{F_2}$ is $\frac{\alpha}{\sqrt{5}}$. The value of α is _____.



(1) 2 (2) 3

(3) 1 (4) $2\sqrt{5}$

PROPERTIES OF SOLIDS

29. Two wires *A* and *B* made of different materials of length 6.0 cm and 5.4 cm, respectively and area of cross sections 3.0×10^{-5} m^2 and 4.5×10^{-5} m^2, respectively are stretched by the same magnitude under a given load. The ratio of the Young's modulus of *A* to that of *B* is x : 3. The value of x is _____.

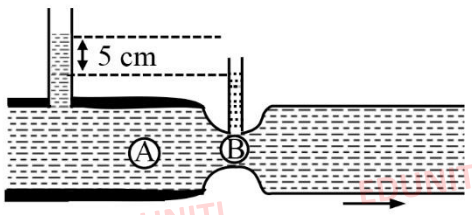
(1) 1 (2) 4

(3) 2 (4) 5

FLUID MECHANICS

30. Water flows through a horizontal tube as shown in the figure. The difference in height between the water columns in vertical tubes is 5 cm and the area of cross-sections at *A* and *B* are 6 cm^2 and 3 cm^2 respectively.

The rate of flow will be_____cm^3/s. (take g = 10 m/s^2)



(1) $\frac{200}{\sqrt{3}}$ (2) $200\sqrt{6}$

(3) $200\sqrt{3}$ (4) $100\sqrt{3}$

CLICK FOR SOLUTION

31. When a part of a straight capillary tube is placed vertically in a liquid, the liquid raises upto certain height h. If the inner radius of the capillary tube, density of the liquid and surface tension of the liquid decrease by 1 % each, then the height of the liquid in the tube will change by _____%.

(1) –1 (2) +3

(3) –3 (4) +1

32. An air bubble of volume 2.9 cm^3 rises from the bottom of a swimming pool of 5 m deep. At the bottom of the pool water temperature is 17 °C. The volume of the bubble when it reaches the surface, where the water temperature is 27 °C, is____ cm^3. (g = 10 m/s^2, density of water = 10^3 kg/m^3, and 1 atm pressure is 10^5 Pa)

(1) 4.2 (2) 2.0

(3) 3.0 (4) 4.5

SIMPLE HARMONIC MOTION

33. The kinetic energy of a simple harmonic oscillator is oscillating with angular frequency of 176 rad/s. The frequency of this simple harmonic oscillator is_____Hz. $\left[\text{take } \pi = \frac{22}{7}\right]$

(1) 14 (2) 88

(3) 28 (4) 176

34. A cylindrical block of mass M and area of cross section A is floating in a liquid of density ρ and with its axis vertical. When depressed a little and released the block starts oscillating. The period of oscillation is_____.

(1) $2\pi\sqrt{\frac{M}{\rho Ag}}$ (3) $\pi\sqrt{\frac{\rho A}{Mg}}$

(2) $\pi\sqrt{\frac{2M}{\rho Ag}}$ (4) $2\pi\sqrt{\frac{\rho A}{Mg}}$

35. A spring of force constant 15 N/m is cut into two pieces. If the ratio of their length is 1:3, then the force constant of smaller piece is____N/m

(1) 15 (2) 20

(3) 60 (4) 45

THERMAL PROPERTIES

36. A gas based geyser heats water flowing at the rate of 5.0 litres per minute from 27°C to 87°C. The rate of consumption of the gas is ____ g/s. (Take heat of combustion of gas = 5.0×10^4 J/g) specific heat capacity of water = 4200 J/kg. °C

(1) 2.1 (2) 4.2

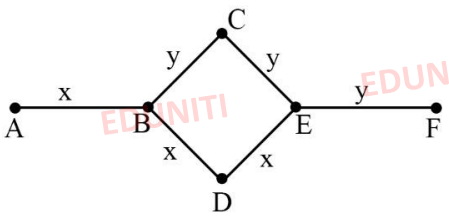
(3) 0.42 (4) 0.21

37. An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at 30°C. The coefficient of linear expansion of aluminium and steel are 24×10^{-6}/°C and 1.2×10^{-5}/°C, respectively. The length of this composite rod when its temperature is raised to 100°C, is _____ cm.

(1) 120.20 (2) 120.15

(3) 120.03 (4) 120.06

38. Rods x and y of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points *A* and *F* are maintained at 100 °C and 40 °C respectively. Given the thermal conductivity of rod x is three times of that of rod y, the temperature at junction points *B* and *E* are (close to) :



(1) 89 °C and 73 °C respectively

(2) 80 °C and 60 °C respectively

(3) 80 °C and 70 °C respectively

(4) 60 °C and 45 °C respectively

39. In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg, moving in opposite directions with speeds of 10 m/s and 30 m/s, respectively, strike each other and stick together. The rise in temperature (in °C), if all the heat produced during the collision is retained by these spheres, is : (specific heat of sphere material 31 cal/kg. °C and 1 cal = 4.2 J)

(1) 1.75 (2) 1.44

(3) 1.15 (4) 1.95

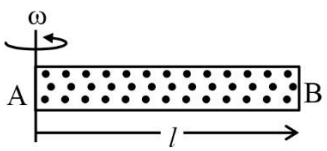
40. A brass wire of length 2m and radius 1mm at 27°C is held taut between two rigid supports. Initially it was cooled to a temperature of –43°C creating a tension T in the wire. The temperature to which the wire has to be cooled in order to increase the tension in it to 1.4T, is ____°C

(1) –86 (2) –71

(3) –65 (4) –80

KTG & THERMODYNAMICS

41. A cylindrical tube AB of length l, closed at both ends contains an ideal gas of 1 mol having molecular weight M. The tube is rotated in a horizontal plane with constant angular velocity ω about an axis perpendicular to AB and passing through the edge at end A, as shown in the figure. If P_A and P_B are the pressures at A and B respectively, then (Consider the temperature is same at all points in the tube)



(1) $P_B = P_A \exp(M\omega^2 l^2 / 2RT)$

(2) $P_B = P_A$

(3) $P_B = P_A \exp(M\omega^2 l^2 / 3RT)$

(4) $P_B = P_A \exp(M\omega^2 l^2 / RT)$

42. The internal energy of a monoatomic gas is 3nRT. One mole of helium is kept in a cylinder having internal cross section area of 17 cm^2 and fitted with a light movable frictionless piston. The gas is heated slowly by suppling 126 J heat. If the temperature rises by 4 °C, then the piston will move_____cm.

(atmospheric pressure = 10^5 Pa)

(1) 14.5 (2) 1.55

(3) 15.5 (4) 1.45

43. Density of water at 4°C and 20°C are 1000 kg/m^3 and 998 kg/m^3 . The increase in internal energy of 4 kg water when it is heated from 4 °C to 20°C is____J.

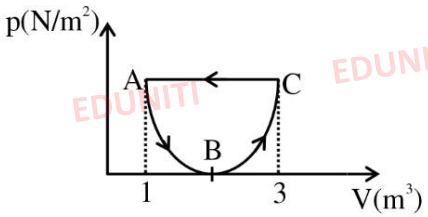
(Specific heat capacity of water = 4200 J/kg°C

1 atmospheric pressure = 10^5 Pa)

(1) 315826.2 (3) 258700.8

(2) 234699.2 (4) 268799.2

44. In the following p-V diagram the equation of state along the curved path is given by $(V-2)^2 = 4ap$ where a is a constant. The total work done in the closed path is



(1) $-\frac{1}{a}$ (2) $+\frac{1}{3a}$

(3) $\frac{1}{2a}$ (4) $-\frac{1}{3a}$

45. The mean free path of a molecule of diameter 5×10^{-10} m at the temperature 41°C and pressure 1.38×10^5 Pa, is given as ____m.

(Given $k_B = 1.38 \times 10^{-23}$ J/K).

(1) $2\sqrt{2} \times 10^{-10}$ (3) $2\sqrt{2} \times 10^{-8}$

(2) $10\sqrt{2} \times 10^{-8}$ (4) 2×10^{-8}

CLICK FOR SOLUTION

WAVE MOTION

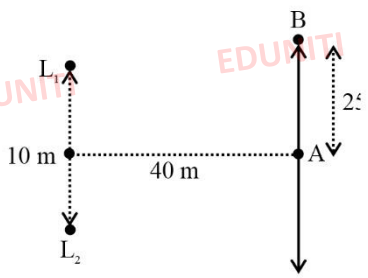
46. Two strings (A, B) having linear densities $\mu_A = 2 \times 10^{-4}$ kg/m and $\mu_B = 4 \times 10^{-4}$ kg/m and lengths $L_A = 2.5$ m and $L_B = 1.5$ m respectively are joined. Free ends of A and B are tied to two rigid supports C and D, respectively creating a tension of 500 N in the wire. Two identical pulses, sent from C and D ends, take time t_1 and t_2, respectively, to reach the joint. The ratio t_1/t_2 is :

(1) 1.08 (2) 1.90

(3) 1.67 (4) 1.18

47. Two loudspeakers (L_1 and L_2) are placed with a separation of 10 m, as shown in figure. Both speakers are fed with an audio input signal of same frequency with constant volume. A voice recorder, initially at point A, at equidistance to both loud speakers, is moved by 25 m along the line AB while monitoring the audio signal. The measured signal was found to undergo 10 cycles of minima and maxima during the movement. The frequency of the input signal is _____ Hz

(Speed of sound in air is 324 m/s and $\sqrt{5} = 2.23$)



48. The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is 5/x. The value of x is_____.

(1) 4 (2) 2 (3) 1 (4) 3

49. The speed of a longitudinal wave in a metallic bar is 400 m/s. If the density and Young's modulus of the bar material are increased by 0.5% and 1% respectively then the speed of the wave is changed approximately to _____ m/s.

(1) 399 (2) 398 (3) 402 (4) 401

50. Two tuning forks A and B are sounded together giving rise to 8 beats in 2 s. When fork A is loaded with wax, the beat frequency is reduced to 4 beats in 2 s. If the original frequency of tuning fork B is 380 Hz, then the original frequency of tuning fork A is _____ Hz.

ELECTROSTATICS

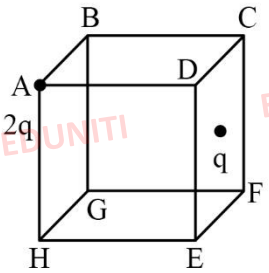
51. Consider two identical metallic spheres of radius R each having charge Q and mass m. Their centers have an initial separation of $4R$. Both the spheres are given an initial speed of u towards each other. The minimum value of u, so that they can just touch each other is :

(Take $k = \frac{1}{4\pi\epsilon_0}$ and assume $kQ^2 > Gm^2$ where G is the Gravitational constant)

(1) $\sqrt{\frac{kQ^2}{4mR}\left(1-\frac{Gm^2}{kQ^2}\right)}$ (2) $\sqrt{\frac{kQ^2}{4mR}\left(1+\frac{Gm^2}{kQ^2}\right)}$

(3) $\sqrt{\frac{kQ^2}{2mR}\left(1-\frac{Gm^2}{kQ^2}\right)}$ (4) $\sqrt{\frac{kQ^2}{2mR}\left(1-\frac{Gm^2}{2kQ^2}\right)}$

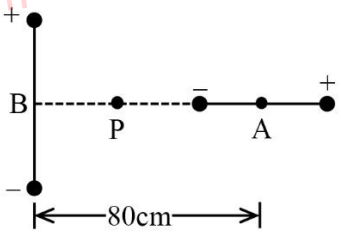
52. Two point charges 2q and q are placed at vertex A and centre of face CDEF of the cube as shown in figure. The electric flux passing through the cube is :



(1) $\frac{3q}{\varepsilon_0}$ (2) $\frac{q}{\varepsilon_0}$ (3) $\frac{3q}{2\varepsilon_0}$ (4) $\frac{3q}{4\varepsilon_0}$

CLICK FOR SOLUTION

53. Two shorts dipoles (A, B), A having charges $\pm$ 2μC and length 1 cm and B having charges $\pm$ 4μC and length 1 cm are placed with their centres 80 cm apart as shown in the figure. The electric field at a point P, equi-distant from the centres of both dipoles is_____N/C.



(1) $\frac{9}{16}\sqrt{2}\times10^5$ (2) $4.5\sqrt{2}\times10^4$

(3) $9\sqrt{2}\times10^4$ (4) $\frac{9}{16}\sqrt{2}\times10^4$

54. Two charges 7μC and –2 μC are placed at (–9, 0, 0) cm and (9, 0, 0) cm respectively in an external field $E = \frac{A}{r^2}\hat{r}$, where $A = 9 \times 10^5$ N/C.m^2.

Considering the potential at infinity is 0, the electrostatic energy of the configuration is_____J.

(1) 1.4 (2) –90.7

(3) 49.3 (4) 24.3

55. The electrostatic potential in a charged spherical region of radius r varies as $V = ar^3 + b$, where a and b are constants. The total charge in the sphere of unit radius is $\alpha \times \pi a \in_0$. The value of α is _____.

(permittivity of vacuum is $\in_0$)

(1) –12 (2) –6

(3) –9 (4) –8

56. There are three co-centric conducting spherical shells A, B and C of radii a, b and c respectively. The potential of the spheres A, B and C respectively, are (c>b>a and charge on A, B and C are q1, q2 and q3 respectively)

(1) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{a}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{b}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{c}\right)$

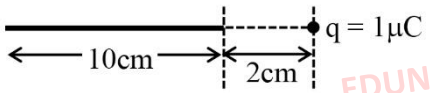
(2) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{a}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right)$

(3) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{c}\right)$

(4) $\frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{a}+\frac{q_2}{b}+\frac{q_3}{c}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{b}\right), \frac{1}{4\pi\epsilon_0}\left(\frac{q_1+q_2+q_3}{c}\right)$

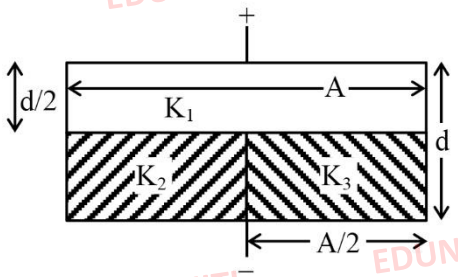
57. A point charge q = 1 μC is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge Q = 24 μC, distributed uniformly along its length, as shown in figure. Force between q and wire is _____ N.

(Use $\frac{1}{4\pi\epsilon_0} = 9\times10^9 \text{N.m}^2/\text{C}^2$)



CAPACITORS

58. The space between the plates of a parallel-plate capacitor of capacitance C (without any dielectric) is now filled with three dielectric slabs of dielectric constants $K_1 = 2$, $K_2 = 3$, and $K_3 = 5$ (as shown in the figure). If new capacitance is $\frac{n}{3}$C then the value of n is_____.

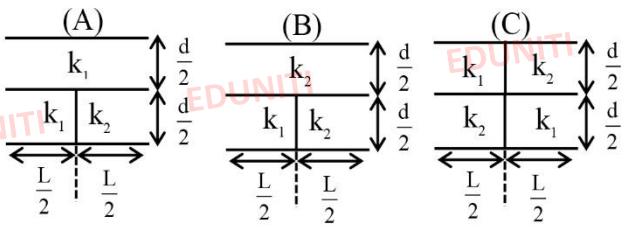


59. A parallel plate capacitor with plate separation 5 mm is charged by a battery. On introducing a mica sheet of 2 mm and maintaining the connections of the plates with the terminals of the battery, it is found that it draws 25% more charge from the battery. The dielectric constant of mica is____.

(1) 2.5 (2) 2.0

(3) 1.5 (4) 1.0

60. Three parallel plate capacitors each with area A and separation d are filled with two dielectric (k_1 and k_2) in the following fashion. Which of the following is true? ($k_1 > k_2$)



(1) $C_B > C_C > C_A$ (2) $C_C > C_B > C_A$

(3) $C_C > C_A > C_B$ (4) $C_A > C_C > C_B$

61. Identify the correct statements :

A. Effective capacitance of a series combination of capacitors is always smaller than the smallest capacitance of the capacitor in the combination.

B. When a dielectric medium is placed between the charged plates of a capacitor, displacement of charges cannot occur due to insulation property of dielectric.

C. Increasing of area of capacitor plate or decreasing of thickness of dielectric is an alternate method to increase the capacitance.

D. For a point charge, concentric spherical shells centered at the location of the charge are equipotential surfaces.

Choose the **correct** answer from the options given below.

(1) A, B and C only (3) A, C and D only

(2) C and D only (4) B and D only

CURRENT ELECTRICITY

62. An electric power line having total resistance of 2 Ω, delivers 1 kW of power of 250 V. The percentage efficiency of transmission line is ____ .

(A) 96.9 (B) 86.5

(C) 100 (D) 92.5

63. A cylindrical conductor of length 2m and area of cross-section 0.2 mm^2 carries an electric current of 1.6 A when its ends are connected to a 2V battery. Mobility of electrons in the conductor is $\alpha \times 10^{-3}$ $m^2/V.s$. The value of α is :

(electron concentration = $5 \times 10^{28}/m^3$ and electron charge = 1.6×10^{-19} C)

64. A regular hexagon is formed by six wires each of resistance r Ω and the corners are joined to centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

(1) $\frac{4}{5}r$ (2) $\frac{5}{8}r$

(3) $\frac{3}{4}r$ (4) $\frac{3}{5}r$

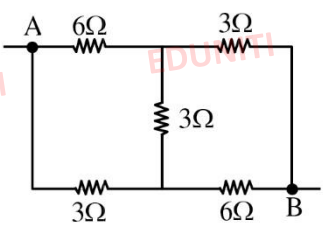
65. In a meter bridge experiment to determine the value of unknown resistance, first the resistances 2 Ω and 3 Ω are connected in the left and right gaps of the bridge and the null point is obtained at a distance l cm from the left. Now when an unknown resistance x Ω is connected in parallel to 3 Ω resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is ___ Ω.

66. In the potentiometer, when the cell in the secondary circuit is shunted with 4 Ω resistance, the balance is obtained at the length 120 cm of wire. Now when the same cell in shunted with 12 Ω resistance, the balance is shifted to a length of 180 cm. The internal resistance of cell is ___ Ω.

(1) 3 (2) 4

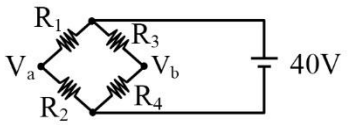
(3) 12 (4) 6

67. The equivalent resistance between the points A and B in the following circuit is $\frac{x}{5}\Omega$. The value of x is _____.



CLICK FOR SOLUTION

68. A wheatstone bridge is initially at room temperature and all arms of the bridge have same value of resistances ($R_1 = R_2 = R_3 = R_4$). When R_3 resistance is heated to some temperature, its resistance value has gone up by 10%. The potential difference ($V_a - V_b$) (after R_3 is heated) is _____V.

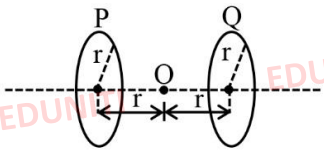


(1) 1.05 (2) 0 (3) 0.95 (4) 2

MOVING CHARGES & MEC

69. A short bar magnet placed with its axis at 30° with an external field of 800 Gauss, experiences a torque of 0.016 N.m. The work done in moving it from most stable to most unstable position is $\alpha \times 10^{-3}$ J. The value of α is _____.

70.

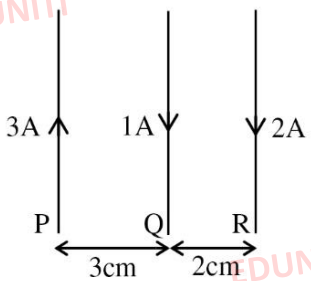


Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and 4I respectively in clockwise direction as seen from O. The net magnetic field at O is:

(1) $\frac{3\mu_0 I}{4\sqrt{2}r}$ toward P (3) $\frac{\mu_0 I}{4\sqrt{2}r}$ towards Q

(2) $\frac{\mu_0 I}{4\sqrt{2}r}$ toward P (4) $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q

71. Three long straight wires carrying current are arranged mutually parallel as shown in the figure. The force experienced by 15 cm length of wire Q is _____.



($\mu_0 = 4\pi \times 10^{-7}$ T.m/A)

(1) 6×10^{-7} N towards P (3) 6×10^{-7} N towards R

(2) 6×10^{-6} N towards R (4) 6×10^{-6} N towards P

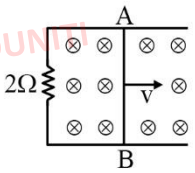
ELECTROMAGNETIC INDUCTION

72. A conducting circular loop of area 1.0 m^2 is placed perpendicular to a magnetic field which varies as B = sin(100 t) Tesla. If the resistance of the loop is 100 Ω, then the average thermal energy dissipated in the loop in one period is _____J.

(1) $\frac{\pi}{2}$ (2) 2π

(3) π (4) π^2

73. A 1 m long metal rod AB completes the circuit as shown in figure. The area of circuit is perpendicular to the magnetic field of 0.10 T. If the resistance of the total circuit is 2Ω then the force needed to move the rod towards right with constant speed (v) of 1.5 m/s is _____ N.



(1) 7.5×10^{-2} (2) 5.7×10^{-3}

(3) 5.7×10^{-2} (4) 7.5×10^{-3}

74. Three identical coils C_1, C_2 and C_3 are closely placed such that they share a common axis. C_2 is exactly midway. C_1 carries current I in anti-clockwise direction while C_3 carries current I in clockwise direction. An induced current flows through C_2 will be in clockwise direction when

(1) C_1 and C_3 move with equal speeds away from C_2

(2) C_1 moves towards C_2 and C_3 moves away from C_2

(3) C_1 moves away from C_2 and C_3 moves towards C_2

(4) C_1 and C_3 move with equal speeds towards C_2

75. Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω. The energy density in the inductor when the current reaches $\left(\frac{1}{e}\right)$ of its maximum value is $\alpha\pi \times \frac{1}{e^2}$ J/m^3. The value of α is ___. ($\mu_0 = 4\pi \times 10^{-7}$ Tm/A).

CLICK FOR SOLUTION

76. A conducting circular loop is rotated about its diameter at a constant angular speed of 100 rad/s in a magnetic field of 0.5T perpendicular to the axis of rotation. When the loop is rotated by 30° from the horizontal position, the induced EMF is 15.4 mV. The radius of the loop is _____ mm.

$\left(\text{Take } \pi = \frac{22}{7}\right)$

77. A simple pendulum made of mass 10 g and a metallic wire of length 10 cm is suspended vertically in a uniform magnetic field of 2 T. The magnetic field direction is perpendicular to the plane of oscillations of the pendulum. If the pendulum is released from an angle of 60° with vertical, then maximum induced EMF between the point of suspension and point of oscillation is _____ mV. (Take g = 10 m/s²)

78. Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current I = 10 sin (ωt) A, where ω = 1000 rad./s. A circular conducting loop (B) of radius 1 cm coaxially slided through the solenoid at a speed v = 1 cm/s. The r.m.s. current through the loop when the coil B is inserted 10 cm inside the solenoid is $\alpha/\sqrt{2}$ μA. The value of α is _____.

[Resistance of the loop = 10 Ω]

(1) 197 (2) 80

(3) 280 (4) 100

CLICK FOR SOLUTION

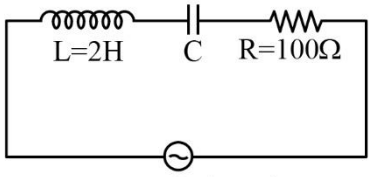
ALTERNATINJG CURRENT

79. A capacitor C is first charged fully with potential difference of V_0 and disconnected from the battery. The charged capacitor is connected across an inductor having inductance L. In t s 25% of the initial energy in the capacitor is transferred to the inductor. The value of t is_____s.

(1) $\frac{\pi\sqrt{LC}}{3}$ (2) $\frac{\pi\sqrt{LC}}{6}$

(3) $\frac{\pi\sqrt{LC}}{2}$ (4) $\pi\sqrt{\frac{LC}{2}}$

80. Using a variable-frequency a.c. voltage source, the maximum current measured in the given LCR circuit is 50 mA for V=5sin(100t). The values of L and R are shown in the figure. The capacitance of the capacitor (C) used is _____ μF.



V=5 sin (100 t)

81. The electric current in the circuit is given as $i = i_0(t/T)$. The r.m.s current for the period t = 0 to $t = T$ is _____

(1) $\frac{i_0}{\sqrt{2}}$ (2) i_0

(3) $\frac{i_0}{\sqrt{6}}$ (4) $\frac{i_0}{\sqrt{3}}$

WAVE OPTICS

82. In a double slit experiment the distance between the slits is 0.1 cm and the screen is placed at 50 cm from the slits plane. When one slit is covered with a transparent sheet having thickness t and refractive index n(= 1.5), the central fringe shifts by 0.2 cm. The value of t is ____ cm.

(1) 8×10^{-4} (2) 6.0×10^{-3}

(3) 5.6×10^{-4} (4) 5.0×10^{-3}

83. When an unpolarized light falls at a particular angle on a glass plate (placed in air), it is observed that the reflected beam is linearly polarized. The angle of refracted beam with respect to the normal is _____.

($\tan^{-1}(1.52) = 57.7°$, refractive indices of air and glass are 1.00 and 1.52, respectively)

(1) 39.6° (2) 32.3°

(3) 42.6° (4) 36.3°

84. In the Young's double slit experiment the intensity produced by each one of the individual slits is I_0. The distance between two slits is 2 mm. The distance of screen from slits is 10 m. The wavelength of light is 6000 Å. The intensity of light on the screen in front of one of the slits is ____.

(1) $2I_0$ (2) I_0 (3) $\frac{I_0}{2}$ (4) $4I_0$

EM WAVES

85. An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma = 10$ mho/m. The ratio of maximum conducting current density to maximum displacement current density is _____.

$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9\times10^9 \text{Nm}^2/\text{C}^2\right]$

86. The electric field of a plane electromagnetic wave, travelling in an unknown non-magnetic medium is given by,

$E_y = 20 \sin(3 \times 10^6 x - 4.5 \times 10^{14} t)$ V/m

(where x, t and other values have S.I. units). The dielectric constant of the medium is ____.

(speed of light in free space is 3×10^8 m/s)

87. A laser beam has intensity of 4.0×10^{14} W/m^2. The amplitude of magnetic field associated with beam is _____ T. (Take $\epsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2 and $c = 3 \times 10^8$ m/s)

(1) 2.0 (2) 18.3

(3) 5.5 (4) 1.83

88. The equation of the electric field of an electromagnetic wave propagating through free space is given by :

$E = \sqrt{377}\ \sin(6.27 \times 10^3 t - 2.09 \times 10^{-5} x)$ N/C

he average power of the electromagnetic wave is $\left(\frac{1}{\alpha}\right)$ W/m^2. The value of α is _____

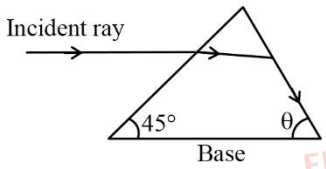
$\left(\text{Take } \sqrt{\frac{\mu_0}{\epsilon_0}} = 377 \text{ in SI units}\right)$

RAY OPTICS

89. A collimated beam of light of diameter 2 mm is propagating along x-axis. The beam is required to be expanded in a collimated beam of diameter 14 mm using a system of two convex lenses. If first lens has focal length 40 mm, then the focal length of second lens is _____ mm.

90. In a microscope the objective is having focal length $f_0 = 2$ cm and eye-piece is having focal length $f_e = 4$ cm. The tube length is 32 cm. The magnification produced by this microscope for normal adjustment is _____.

91. As shown in the diagram, when the incident ray is parallel to base of the prism, the emergent ray grazes along the second surface.



If refractive index of the material of prism is $\sqrt{2}$, the angle θ of prism is.

(1) 60° (2) 75°

(3) 90° (4) 45°

CLICK FOR SOLUTION

92. A thin convex lens of focal length 5 cm and a thin concave lens of focal length 4 cm are combined together (without any gap) and this combination has magnification m_1 when an object is placed 10 cm before the convex lens. Keeping the positions of convex lens and object undisturbed a gap of 1 cm is introduced between the lenses by moving the concave lens away, which lead to a change in magnification of total lens system to m_2.

The value of $\left|\frac{m_1}{m_2}\right|$ is _____.

(1) $\frac{5}{9}$ (2) $\frac{5}{27}$

(3) $\frac{3}{2}$ (4) $\frac{25}{27}$

93. A thin prism with angle 5° of refractive index 1.72 is combined with another prism of refractive index 1.9 to produce dispersion without deviation. The angle of second prism is _____.

(1) 4.5° (2) 6°

(3) 4° (4) 5°

94. A prism of angle 75° and refractive index $\sqrt{3}$ is coated with thin film of refractive index 1.5 only at the back exit surface. To have total internal reflection at the back exit surface the incident angle must be _____.

(sin 15° = 0.25 and sin 25° = 0.43)

(1) between 15° and 20° (2) 15°

(3) > 25° (4) < 15°

95. The size of the images of an object, formed by a thin lens are equal when the object is placed at two different positions 8 cm and 24 cm from the lens. The focal length of the lens is _____ cm.

96. Distance between an object and three times magnified real image is 40 cm. The focal length of the mirror used is _____ cm.

(1) – 15/2 (2) – 10

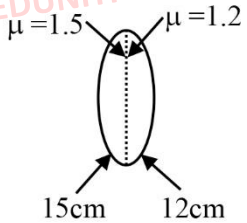
(3) – 20 (4) – 15

97. The magnitudes of power of a biconvex lens (refractive index 1.5) and that of a plano-concave lens (refractive index = 1.7) are same. If the curvature of plano-concave lens exactly matches with the curvature of back surface of the biconvex lens, then ratio of radius of curvature of front and back surface of the biconvex lens is _____ .

(1) 5 : 2 (2) 5 : 12

(3) 12 : 5 (4) 2 : 5

98. A biconvex lens is formed by using two thin planoconvex lenses, as shown in the figure. The refractive index and radius of curved surfaces are also mentioned in figure. When an object is placed on the left side of lens at a distance of 30 cm from the biconvex lens, the magnification of the image will be :



(1) –2 (2) +2

(3) +2.5 (4) –2.5

99. For a transparent prism, if the angle of minimum deviation is equal to its refracting angle, the refractive index n of the prism satisfies.

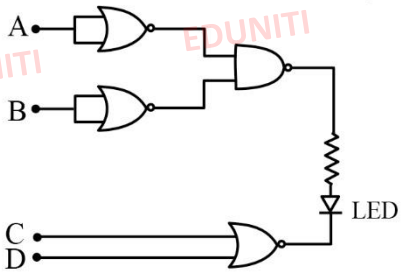
(1) $\sqrt{2} < n < 2\sqrt{2}$ (2) $1 < n < 2$

(3) $n \geq 2$ (4) $\sqrt{2} < n < 2$

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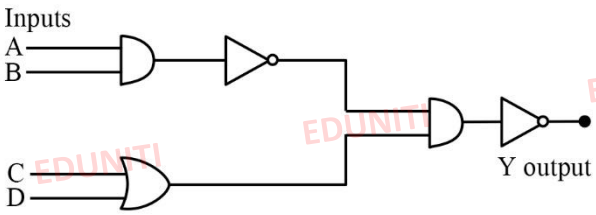
SEMICONDUCTORS

100. Find the correct combination of A, B, C and D inputs which can cause the LED to glow.



(1) 0100 (2) 0011
(3) 1000 (4) 1101

101. The correct truth table for the given input data of the following logic gate is :



(1)

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	1
1	1	1	1	0

(2)

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	0
1	1	1	1	1

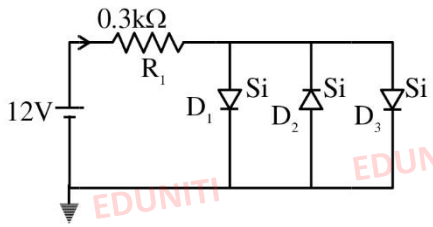
(3)

Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	0
1	0	1	0	1
1	1	1	1	1

(4)

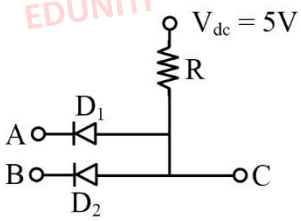
Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	1
1	0	1	0	1
1	1	1	1	1

102. Assuming in forward bias condition there is a voltage drop of 0.7 V across a silicon diode, the current through diode D_1 in the circuit is _____ mA. (Assume all diodes in the given circuit are identical)



(1) 20.15 (2) 11.7
(3) 17.6 (4) 18.8

103. Two p-n junction diodes D_1 and D_2 are connected as shown in figure. A and B are input signals and C is the output. The given circuit will function as a ____.



(1) OR Gate (2) NOR Gate
(3) NAND Gate (4) AND Gate

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MODERN PHYSICS

104. A light wave described by $E = 60[\sin(3 \times 10^{15})t + \sin(12 \times 10^{15})t]$ (in SI units) falls on a metal surface of work function 2.8 eV. The maximum kinetic energy of ejected photoelectron is (approximately) _____ eV. ($h = 6.6 \times 10^{-34}$ J-s. and $e = 1.6 \times 10^{-19}$C)

(1) 5.1 (2) 3.8

(3) 6.0 (4) 7.8

105. If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is _____ m.

(Atomic number of gold = 79 and $\frac{1}{4\pi \in_0} = 9 \times 10^9$ in SI units)

(1) 2.95×10^{-14} (2) 2.95×10^{-16}

(3) 3.85×10^{-16} (4) 3.85×10^{-14}

106. The energy of an electron in an orbit of the Bohr's atom is $-0.04E_0$eV where E_0 is the ground state energy. If L is the angular momentum of the electron in this orbit and h is the Planck's constant, then $\frac{2\pi L}{h}$ is _____ :

(1) 2 (2) 4

(3) 5 (4) 6

107. The minimum frequency of photon required to break a particle of mass 15.348 amu into 4α particles is ____ kHz.

[mass of He nucleus = 4.002 amu,

1 amu = 1.66×10^{-27} kg, $h = 6.6 \times 10^{-34}$ J.s and $c = 3 \times 10^8$ m/s]

(1) 9×10^{19}

(2) 9×10^{20}

(3) 14.94×10^{20}

(4) 14.94×10^{19}

108. The de Broglie wavelength of an oxygen molecule at 27°C is $x \times 10^{-12}$ m. The value of x is (take Planck's constant = 6.63×10^{-34} J.s, Boltzmann constant = 1.38×10^{-23} J/K, mass of oxygen. Molecule = 5.31×10^{-26} kg).

(1) 26 (2) 24

(3) 30 (4) 20

109. In hydrogen atom spectrum, (R → Rydberg's constant)

A. the maximum wavelength of the radiation of Lyman series is $\frac{4}{3R}$

B. the Balmer series lies in the visible region of the spectrum

C. the minimum wavelength of the radiation of Paschen series is $\frac{9}{R}$

D. the minimum wavelength of Lyman series is $\frac{5}{4R}$

Choose the **correct** answer from the options given below :

(1) B, D Only (2) A, B and C Only

(3) A, B and D Only (4) A, B Only

110. The binding energy for the following nuclear reactions are expressed in MeV.

$_2He^3 + {}_0n^1 \rightarrow {}_2He^4 + 20$ MeV

$_2He^4 + {}_0n^1 \rightarrow {}_2He^5 - 0.9$ MeV

If X_3, X_4, X_5 denote the stability of $_2He^3$, $_2He^4$ and $_2He^5$, respectively, then the correct order is :

(1) $X_4 > X_5 > X_3$

(2) $X_4 = X_5 = X_3$

(3) $X_4 > X_5 < X_3$

(4) $X_4 < X_5 < X_3$

111. A nucleus has mass number α and radius R_α. Another nucleus has mass number β and radius R_β. If $\beta = 8\alpha$ then R_α/R_β is :

(1) 2 (2) 8

(3) 1 (4) 0.5

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FILL YOUR ANSWER

Q	ANS
1.	
2.	
3.	
4.	
5.	
6.	
7.	
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Q	ANS
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Q	ANS
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Q	ANS
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103.	
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105.	
106.	
107.	
108.	
109.	
110.	
111.	

CLICK FOR SOLUTION

ANSWER KEY

Q	ANS
1.	4
2.	3
3.	3
4.	3
5.	1
6.	2
7.	4
8.	3
9.	2
10.	4
11.	3
12.	2
13.	210
14.	1
15.	2
16.	4
17.	2
18.	2
19.	2
20.	2
21.	3
22.	1
23.	109
24.	1
25.	4
26.	2
27.	4
28.	1
29.	4
30.	3
31	4

Q	ANS
32.	4
33.	1
34.	1
35.	3
36.	3
37.	2
38.	1
39.	2
40.	2
41.	1
42.	3
43.	4
44.	4
45.	3
46.	4
47.	600
48.	2
49.	4
50.	384
51.	1
52.	4
53.	4
54.	3
55.	1
56.	3
57.	90
58.	8
59.	2
60.	4
61.	3
62.	A

Q	ANS
63.	1
64.	1
65.	6
66.	2
67.	21
68.	3
69.	64
70.	4
71.	2
72.	3
73.	4
74.	2
75.	20 *(length of inductor is missing, answer comes if you take n and N as same)*
76.	14
77.	100
78.	1
79.	2
80.	50
81.	4
82.	1
83.	2
84.	2
85.	1800
86.	4
87.	4
88.	2
89.	280
90.	100
91.	1
92.	Dropped *(5/6 is correct answer)*
93.	3

Q	ANS
94.	1,2,4
95.	16
96.	4
97.	1
98.	1
99.	4
100.	4
101.	2
102.	4
103.	4
104.	1
105.	1
106.	3
107.	4
108.	1
109.	2
110.	1
111.	4

CLICK FOR SOLUTION